

MA201BS: MATHEMATICS – II
(ADVANCED CALCULUS& ELEMENTARY
COMBINATORICS) (Common to CE, EEE, ME, ECE, CSE, IT,
CSE(IoT), AI&ML, DSE)

B.Tech. I Year II Semester								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
MA201BS	BSC	L	T	P	C	CIA	SEE	Total
		3	1	-	4	30	70	100
Contact Classes: 45	Tutorial Classes: 15	Practical Classes: Nil				Total Classes: 60		
Prerequisite: Mathematical Knowledge of 12 th / Intermediate level								
Course Objectives: To learn • Methods of solving the differential equations of first and higher order. • Evaluation of multiple integrals and their applications • The physical quantities involved in engineering field related to vector valued functions • The basic properties of vector valued functions and their applications to line, surface and volume integrals • Concept of Recurrence Relations and generating functions								
Course Outcomes: After learning the contents of this paper the student must be able to • Identify whether the given differential equation of first order is exact or not • Solve higher differential equation and apply the concept of differential equation to real world problems • Evaluate the multiple integrals and apply the concept to find areas, volumes, centre of mass and Gravity for cubes, sphere and rectangular parallelepiped • Evaluate the line, surface and volume integrals and converting them from one to another • Apply the concepts of advanced counting techniques								
Unit: 1	First Order ODE							
Exact, linear and Bernoulli's equations; Applications : Newton's law of cooling, Law of natural growth and decay; Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type.								
Unit: 2	Ordinary Differential Equations of Higher Order							
Second order linear differential equations with constant coefficients: Non-Homogeneous terms of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x, $e^{ax}V(x)$ and $x V(x)$; method of variation of parameters; Equations reducible to linear ODE with constant coefficients: Legendre's equation, Cauchy-Euler equation.								
Unit: 3	Multivariable Calculus (Integration)							
Evaluation of Double Integrals (Cartesian and polar coordinates); change of order of integration (only Cartesian form); Evaluation of Triple Integrals: Change of variables (Cartesian to polar) for double and (Cartesian to Spherical and Cylindrical polar coordinates) for triple integrals. Applications: Areas (by double integrals) and volumes (by double integrals and triple integrals), Centre of mass and Gravity (constant and variable densities) by double and triple integrals (applications involving cubes, sphere and rectangular parallelepiped).								
Unit: 4	Vector calculus							
Gradient, Divergence, Curl, Line integral, conservative fields, Green's theorem, surface area of solids of revolution, surface area, surface integral, Triple integrals and Gauss Divergence theorem, Stokes' theorem (without proofs)								
Unit: 5	Counting (Recurrence Relations & Elementary Combinatorics)							
Basic counting, Pigeonhole principle, Permutations and Combinations, Binomial Coefficients, Application of Recurrence Relations, Solution of Recurrence Relations, Generating functions, Inclusion – Exclusion and applications								

Text Books:

1. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers, 36th Edition, 2010
2. Kenneth H. Rosen, Discrete Mathematics and Its Applications, McGrawHill.

Reference Books:

1. S. L. Ross, Differential Equations, 3rd Ed., Wiley India, 1984.
2. G.B. Thomas and R.L. Finney, Calculus and Analytic geometry, 9th Edition, Pearson, Reprint, 2002.
3. Higher Engineering Mathematics by B V Ramana, Tata McGraw-Hill
4. Discrete Mathematics for Computer Scientists and Mathematicians by Joe R. Mott, Abraham Kandel, Theodore P. Baker, Prentice-Hall of India Pvt.Ltd.

Web References:

- 1) SWAYAM Online Courses <https://storage.googleapis.com/uniquecourses/online.html>
- 2) Directory of Open Access Journals <https://doaj.org/>
- 3) Springer Open Journals <https://www.springeropen.com/journals>
- 4) UG/PG MOOCs http://ugcmoocs.inflibnet.ac.in/ugcmoocs/moocs_courses.php

E-Text Books:

- 1) National Digital Library: <https://ndl.iitkgp.ac.in/>
- 2) NCERT Text Books <http://ncert.nic.in/textbook/textbook.htm>
- 3) Directory of Open Access Books <https://www.doabooks.org/>

